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Paleomagnetic and Structural Constraints on Rotations in the North Chilean Coast Ranges

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Abstract— Paleomagnetic analysis of seventy-six samples from six sites in the Jurassic to Cretaceous plutons (radiometric ages from 157 to 128 Ma) of the Coastal Cordillera of northern Chile yielded stable remanent magnetization directions. The plutons record both a normal and a reversed characteristic direction. The mean normal direction has a declination of 3.4°, an inclination of -38.5°, and an alpha 95 of 10.5°. The mean reversed direction has a declination of 207.8°, an inclination of 33.8°, and an alpha 95 of 10.8°. Paleopoles calculated from these mean directions have the positions of latitude=85.8° N, longitude=338.2° E and latitude=63.5° S, longitude=191.9° E respectively. In comparison to an APW path, the reversed pole indicates a clockwise rotation of approximately 28°, similar to other regional results, whereas the normal pole is concordant with the present day magnetic field. These results, in and of themselves, could support oroclinal bending or local block rotations. However arrays of secondary faults to the Atacama fault system in the sampling area show characteristic spacings, orientations, and displacements. Restoration of displacements results in a 31° clockwise rotation of fault blocks; essentially removing the observed paleomagnetic discordance. Thus, structural data confirms the local block rotation model. The existence of clockwise discordant paleomagnetic data within the sinistral (counterclockwise) Atacama Fault Zone (AFZ) underscores the ambiguity of using the sense of rotation to infer the sense of motion on the larger shear zone. Age constraints for motions along the AFZ suggest that these rotations were completed in the Early Cretaceous; essentially coeval with magmatic activity within the Coast Ranges, and underscores the magmatic arc, rather than forearc, setting for this tectonic event.

Resumen—Análisis paleomagnéticos de setenta y seis muestras provenientes de seis diferentes lugares en los plutones Jurásico-Cretácico (157 - 128 Ma) de la cordillera costera de el norte de Chile, permitio establecer las direcciones de el magnetismo remanente.Las direcciones registradas por los plutones son tanto normal como inversa. La dirección normal media tiene una declinación de 3.4∞, una inclinación de -38.5° y un alpha 95 de 10.5°. La dirección reversa media tiene una declinación de 207.8° una inclinación de 33.8° y un alpha 95 de 10.8°. Los paleopolos calculados a partir de estas direcciones medias tienen las posiciones de latitud 85.8° N, longitud 338.2° E y latitud 63.5°S, longitud 191.9°E respectivamente. En comparación a un modelo APW el polo inverso indica una rotación en el sentido de el reloj de aproximadamente 28°, similar a otros resultados regionales, mientras el polo normal es concordante con el campo magnético presente. Estos resultados soportan el doblamiento oroclinal o la rotación local de bloques. Las fallas secundarias de el sistema de fallas de Atacama en el area de estudio muestra un espaciamiento, orientación y desplazamiento caracteristicos. La restauración de desplazamientos resulta en una rotación de 31°, removiendo la discordancia paleomagnetica observada. Asi datos estructurales confirman un modelo de rotación local de bloques. La existencia de datos paleomagnéticos discordantes con el sistema sinestral de fallas de Atacama (AFZ) pone en evidencia la ambigüedad en el uso de el sentido de rotación para inferir el sentido de movimiento en grandes zonas de cizallamiento. La edad para los movimientos de AFZ sugieren que estas rotaciones terminaron en el Cretácico temprano, contemporaneo con actividad magmática dentro de la cordillera de la costa, y señala un arco magmático más que una zona de frente de arco como el marco tectóni para este evento.

INTRODUCTION

THIS PAPER PRESENTS the results of a paleomagnetic investigation of Jurassic to Cretaceous plutons exposed in the Coastal Ranges of Northern Chile between latitudes of 24° to 25° S (Fig.1). It is the fourth paleomagnetic study done in the area, but it is the first to provide results from a regionally extensive sample suite within the Jurassic to Cretaceous coastal batholith.

The objective was to further the understanding of the Mesozoic and Cenozoic tectonic development of the Coastal Cordillera. In recent years, a number of paleomagnetic studies have revealed a complex pattern of post-Paleozoic rotations within the coastal province of Peru and Chile (for review see Beck, 1988; Forsythe *et al.*, 1987). Some workers, in viewing these rotations, have invoked models of oroclinal bending (Heki *et al.*, 1983, 1984; Kono *et al.*, 1985; May and Butler, 1985) following ideas initially expressed by Carey (1955). Others have promoted models of more localized rotations of fault-bounded blocks in the coastal areas as a response to oblique subduction (Beck, 1986, 1988; Hartley *et al.*, 1988), to an "indenter" effect caused by the subduction of anomalous seafloor (Forsythe and Nelson, 1985), or perhaps to some combination of these sub-

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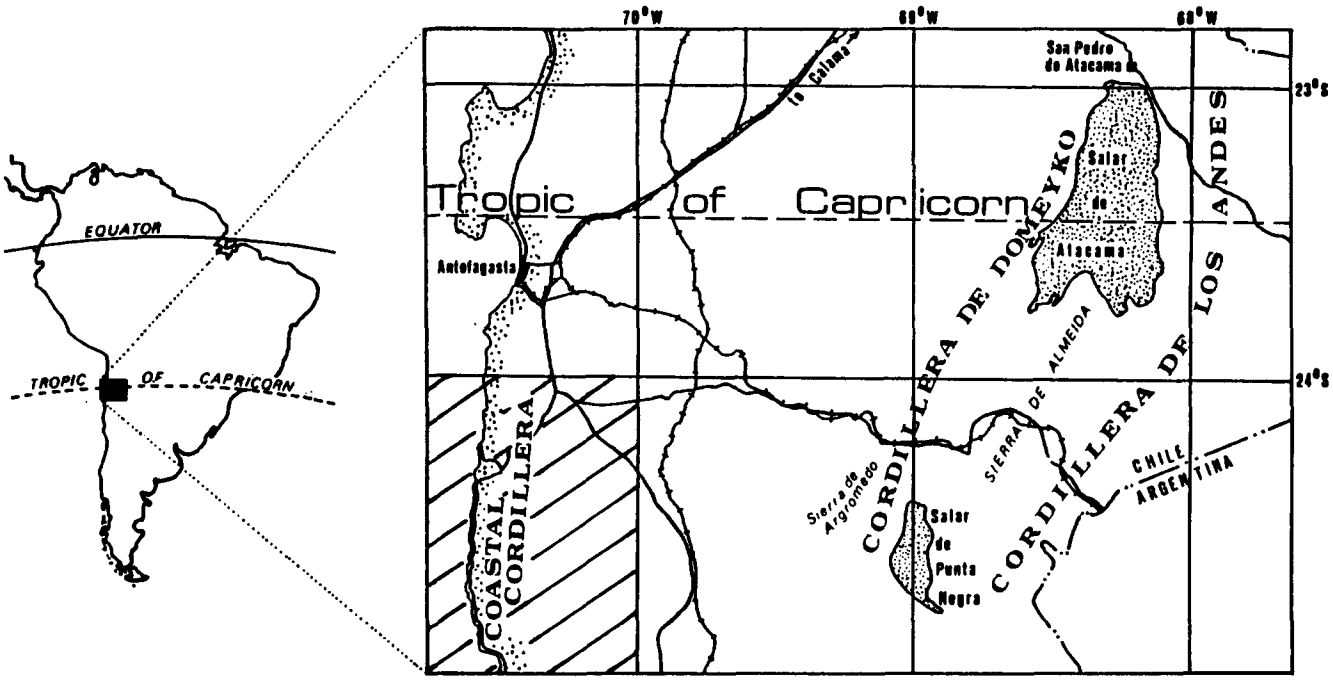


Fig. 1. Location map of South America, hatched section shows general collecting location.

duction-related effects (Garcia *et al.*, 1988). Three coastal studies, Tumer *et al.* (1984), Forsythe *et al.* (1987), and Hartley *et al.* (1988), have been completed in the Antofagasta Province where the Coastal Ranges are separated from a longitudinal depression (Pampa de Taramugal) and the main cordilleran ranges by the Atacama Fault Zone. These studies all yielded data with discordant declinations, suggesting clockwise rotations up to -30°, but had very restricted sampling areas covering a maximum diameter of -10 km. Because of the restricted sampling areas, it has been difficult to ascertain whether the inferred tectonic rotations at these sites are localized or representative of a more regional phenomena such as motion along, or related to, the Atacama Fault Zone. This study was designed to determine the regional extent and character of these inferred rotations. It was aided by the dating in recent years of a suite of the Jurassic and Cretaceous plutons, but is also hindered by the paucity of good outcrops and the general difficulty of access.

GEOLOGIC BACKGROUND

The geology of the sample area is shown in a simplified manner in Figure 2. The oldest unit, "Paleozoic Basement", is a generic name for a collection of plutonic and metamorphic rocks that represent various elements of the region's pre-Jurassic history (see Dalziel and Forsythe, 1985; Coira *et al*, 1982; Bahlburg *et al.*, 1987). This basement is intruded by the "Mesozoic Coastal Batholith" which is composed of gabbros, granodiorites, granites, late-stage aplites, and various mafic to intermediate dike swarms (Garcia, 1967; Ferraris and DiBiase, 1978) that in the sampling area range in age from 128 to 157 Ma (Hervé *et al.*, 1985; see Table 1). Mesozoic sedimentary and volcanic units are the: La Negra Formation (Jurassic volcanics; Garcia, 1967), El Way Formation (Hauterivian marine carbonates; Brueggen, 1950; Wetzel, 1927), and Coloso Formation (Late Cretaceous red beds; Brueggen, 1950). These are grouped in Figure 2 under the label of "Mesozoic Sedimentary and Volcanic Rocks." Unconformably overlying the older units are Tertiary and Quaternary aeolian and shallow marine sandstones and gravels (Feffaris and DiBiase, 1978), which are shown collectively as "Cenozoic Rocks and Unconsolidated Cover." Structurally the region is broken up by N-S and NW trending faults, that, while poorly exposed, are inferred to be high angle (Ferraris and DiBiase, 1978) and are regionally part of the Atacama Fault Zone (discussed below).

The intrusive units of the "Mesozoic Coastal Batholith" and Jurassic to Early Cretaceous volcanic and sedimentary units are interpreted to represent a magmatic arc, the "La Negra Arc" and an associated ensialic back arc basin, respectively. This magmatic arc and back arc basin were developed within the region's basement, essentially along the line of the present day coast ranges. The radiometric ages of rocks resulting from this event range from 190 to 115 Ma. (Maksaev, 1984; Boric *et al*, 1985).

Magmatic activity ceased in the La Negra arc during the Middle Cretaceous, and during the late Cretaceous to early Tertiary the whole Andean domain was deformed in what was referred to as the "Peruana" or "Subhercinica" tectonic phase (Vicente *et al.*, 1973; Davidson, 1984).

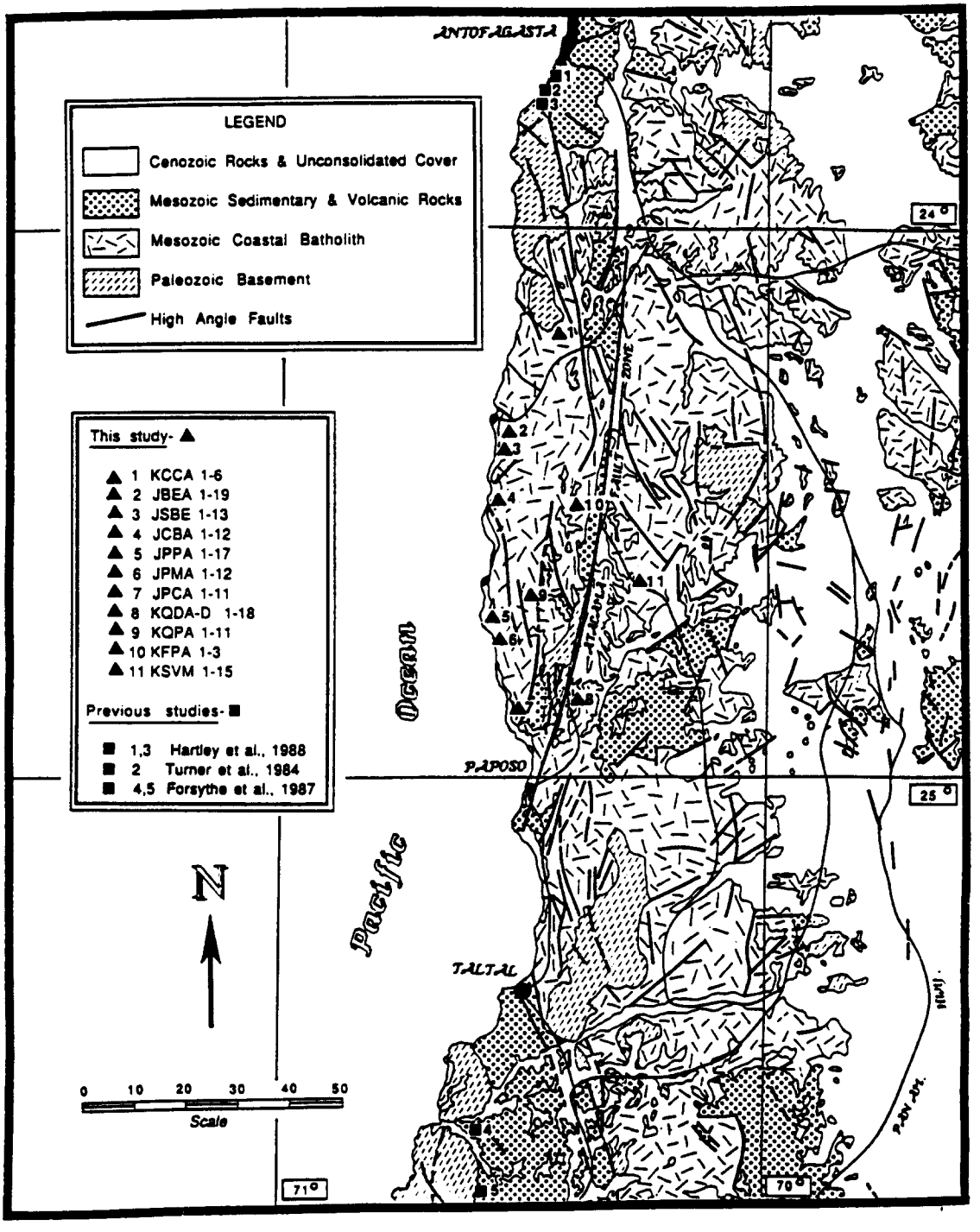


Fig. 2. Simplified geologic map with site locations of this study as well as the locations of other regional studies (after the 1: 1,000,000 scale geological map of Chile, Servicio National de Geologia y Mineria, 1982).

During the latest Cretaceous, the axis of volcanism moved eastward and became established between the Coastal Ranges and the proto-Cordillera Domeyko (Fig. 3). This volcanism lasted until the Eocene, and is represented by the Augusta Victoria and Azabache Formations (Feffaris and DiBiase, 1978). There is evidence of some magmatic activity occurring in the Cordillera de Domeyko during the Oligocene. Here, the north-south alignment of the large porphyry coppers of the Atacama Desert have been dated from 48 to 28 Ma (Boric *et al*, 1985) and appear to have been emplaced through a younger and more eastern longitudinal fault than the Atacama Fault Zone. This younger magmatism is thought to correlate with a period of highly oblique convergence between the Nazca and South American plates (Cande, 1983).

During the Late Oligocene-Early Miocene, there was a major new phase of deformation. The axis of volcanism continued its eastward migration into the Western or present Andean Cordillera in early Miocene time.

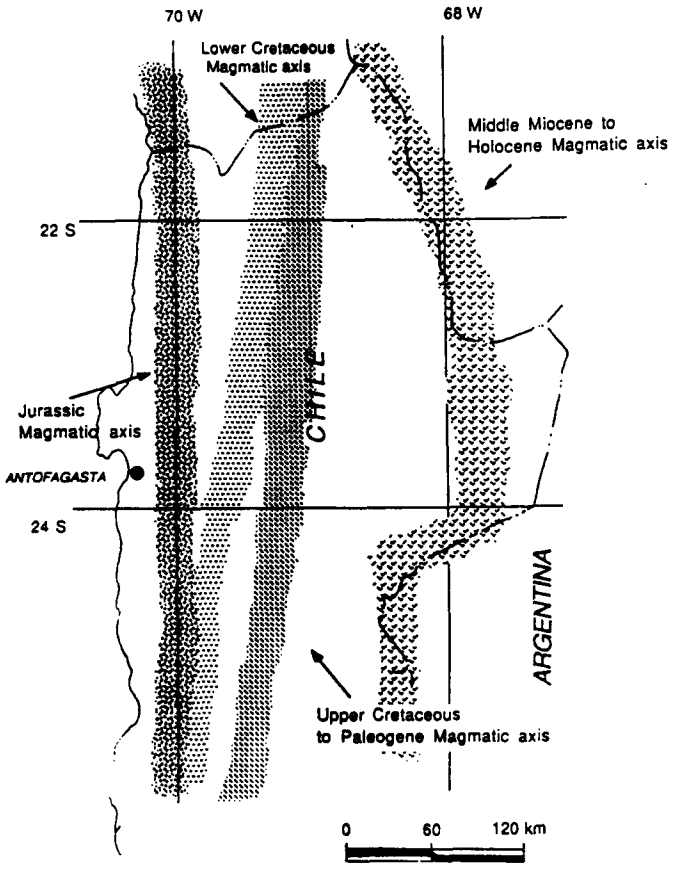


Fig. 3. Map view of the axes of magmatism in northern Chile, dating from the Jurassic until the present (from Maksaev, 1984).

This progressive eastward migration through time is illustrated in Figure 3.

The specific structural history of the Coastal Cordillera in the region of the sampling sites is dominated by its history of high angle block faulting. This is conceptually linked by most researchers to the history of the Atacama Fault Zone. In addition to the faulting, the Mesozoic bedded sequences are also affected in places by gentle folds with NNW-SSE axial trends (Ferraris and DiBiase, 1978). For the most part, however, these sequences in the eastern part of the Coastal Cordillera have a monoclinal structure with N-S strikes and very gentle westward dips.

The Atacama Fault Zone represents one of three major arc-trench parallel fault systems in southern South America (Dalziel and Forsythe, 1985). The Atacama fault system extends for over 800 kilometers from 20° S to the latitude of Copiapo. Over this length it is broken up into three curvilinear segments that form large gently curving oceanward-convex patterns of fault scarps (Arabasz, 1971).

The fault zone is believed to have played an important, if not dominant, role during the late Tertiary and Quaternary in the morphologic development of the Atacama Desert (Mortimer, 1980). During this time period its primary role was to vertically displace the Coastal Cordillera, cutting off drainage patterns that once provided outlets directly to the sea (Mortimer and Rendic, 1975). This late stage history of faulting has been generally assumed to be of normal fault character (Ferrasis and DiBiase, 1978). However, earlier work on the fault (Allen, 1965; Arabasz, 1971) held that it represented a major transcurrent fault system. These earlier studies suggested that the fault system had in part accommodated dextral movements.

More recent mapping of the fault system, combined with the structural analysis of its associated mylonites, within the central (Hervé, 1987b; Scheuber and Andriessen, 1990) and southern (Naranjo *et al.*, 1984) segments now clearly demonstrates that the fault system in these areas has undergone early (Cretaceous) sinistral displacements and then late (late Tertiary) normal fault movements. The central of the three segments is the one that extends from Antofogasta to Taltal, and is directly involved with the units sampled in this study. Here, in this central segment, the youngest pluton affected by sinistral movements is middle Cretaceous in age (144 Ma), and mylonites with left lateral shear sense indicators (Scheuber and Andriessen, 1990) are intruded by undeformed syenogranites of Late Cretaceous age (130 Ma). The same timing has been also determined by Naranjo and others (1984) from their study of the mylonites along the AFZ south of the sampling area which are cut by late stage veins that have been dated by K/Ar to be Neocomian in age. Also, the fault in this southern area is the locus of iron mineralization that has been stratigraphically constrained to be Cretaceous to early Tertiary in age (Thiele and Pindreira, 1984).

Late normal faulting in the central segment has also been further constrained in recent work by Hervé (1987a). Normal faulting is perhaps best exemplified by the main topographic expression of the Atacama fault system within its central segment, which is represented by the Paposo-Salar de Carmen Fault. Here, the coast ranges have been uplifted 300 meters relative to the Pampa de Taramugal to the east. The normal fault motion has cut off long-lived Tertiary drainage networks that previously carried sediment from Sierra Vicuna MacKenna directly westward to the Pacific Ocean. Today these sediments are trapped in a series of playas on the down thrown side of the fault. Tuffs have been dated within one of the uplifted and abandoned river beds, as well as from one of the debris aprons developed on the east flanks of the fault scarp. These dates have constrained the normal displacements between 19 Ma and 5.5 Ma (Hervé, 1987a). None of the drainage networks are displaced horizontally. Thus, it is now fairly clear that:

> of the Atacama fault system, the main topographic expression was a product of late Tertiary normal faulting;
>
> it largely displaced the coastal ranges upwards relative to the lands to the east;
>
> and its history as a transcurrent fault is largely as a sinistral one essentially comagmatic with the

Late Jurassic to Cretaceous development of the coastal batholith.

The older fault history is recorded in a poorly exposed series of mylonites within largely Jurassic plutonic units (Scheuber and Andriessen, 1990), as well as supracrustal faults that have cut and displaced individual Cretaceous plutons along trends oblique to the main fault system by as much as 30 kilometers (Hervé, 1987).

SAMPLING LOCALITIES AND PALEOMAGNETIC TECHNIQUES

The sampling sites were in Jurassic to Cretaceous plutons located in the Coastal Ranges of the Andes in Northern Chile, just south of Antofagasta, between latitudes 24° and 25° S (Fig. 2). The sampling sites are located on both sides of the Atacama Fault Zone. A gas powered drill was used to collect field cores approximately 2.5 cm. in diameter. Individual cores were oriented with an aluminum orienting device to which a Brunton compass was attached. Field checking of local magnetic sources was conducted by periodic surveys of known geographic reference points. In the lab, weathered ends were removed with a rock saw and at least one specimen approximately 2.2 cm. long was cut from the original specimen.

The Natural Remanent Magnetizations (NRMs) of the specimens were measured and then the samples underwent a progressive alternating field (A.F.) demagnetization using the Schonstedt GSD-1 AC Geophysical Specimen Demagnetizer located at Rutgers University, New Brunswick. Samples were treated with A.F. demagnetization in 10 to 12 steps (from 5 to 90 mT). A flux-gate Molyspin spinner magnetometer, also located at Rutgers University, was used for all magnetic measurements. A few samples also underwent additional demagnetization using a Schonstedt TSD-I Thermal Specimen Demagnetizer consisting of nine steps from 100 to 625° C.

After all cores had undergone step-wise demagnetization treatments, the results were encoded for computer aided vectorial and statistical analysis. Individual demagnetization data sets for each sample were then plotted on "Zijderveld" diagrams (Zijderveld, 1967). The plots of each sample were then visually inspected to isolate demagnetization segments which were in turn subjected to a three dimensional least squares, best straight line analysis. This was accomplished using an interactive computer program on the IBM PC adapted by Chris Jesinkey following original work by Kirschvink (1980). This general approach helped to locate and statistically define linear regions (i.e. individual magnetic components) of the data. By grouping common components from each sample within a site, mean directions were then calculated for each site according to standard paleomagnetic procedures (Fisher, 1953). After determining the means and Fisherian parameters, including a precision parameter (k) and an accuracy estimation (alpha 95) for each site, the means were used to calculate average directions for the region as a whole. Paleopoles were calculated from the overall mean declinations and inclinations using the axial geocentric dipole hypothesis.

PALEOMAGNETIC RESULTS

Paleomagnetic laboratory analysis was conducted on 133 samples obtained at 11 localities in the Jurassic to Cretaceous coastal batholith. However, only 6 of the 11 localities (sites JBEA, JSBE, JPPA, JPMA, JPCA and KQDA-D shown in Fig. 2) had a sufficient number of samples (76) to be able to statistically define characteristic directions. Only the results from these six localities are discussed below in any detail. The remainder of the samples were either unstable under demagnetization, formed isolated stable results within otherwise unstable populations, or were found to provide no consistent directions at the site level. Five of the six localities from which stable results were obtained in this study had been previously sampled for radiometric dating (see Table 1), and range in age from 157±4 (Rb/Sr) to 128±3 (K/Ar). Results from each of these six localities are briefly described below, illustrated in Figures 4 - 6, and summarized in Table 1.

The observed demagnetization characteristics permit a grouping of the sites into three populations with distinctive characteristics. While, in general, there were two common characteristic components (normal and reversed) observed in each of the six sites, demagnetization behaviors and the observed directions seen at sites KQDA-D and JPCA warrant a separate discussion for each of their results. The remaining four sites (JBEA, JSBE, JPPA, JPMA) were relatively similar in their characteristics.

The samples from sites JBEA, JSBE, JPPA, and JPMA were all obtained in Jurassic plutons west of the Paposo fault. At each of these sites some cores were encountered with either reversed or normal univectorial magnetizations, as well as some cores with multicomponent remanent magnetizations. The latter samples had variably overlapping coercivity spectra for the normal and reversed components. The demagnetization curves shown in Figures 4 and 5 illustrate the typical pattern seen at these four sites for multicomponent samples, with the normal component being the low coercivity component (removed between 5 and 30 mT). A graph is also shown in Figure 6 of the coercivity spectra encountered for each of the isolated components from all the cores. It further demonstrates that the reversed magnetic component has, in general, a higher coercivity than the normal component. The normal component is very similar to the expected present day field, while the reversed component has a declination rotated significantly clockwise from the antipode of the present day field.

Site JPCA yielded well defined southerly and down directions in the 25 to 90 mT range in all eleven of the collected samples; three of these also had good northerly and up directions in the 5 to 30 mT range. However, the magnetic intensities of a few samples remained relatively high after A.F. cleaning up to 90 mT though

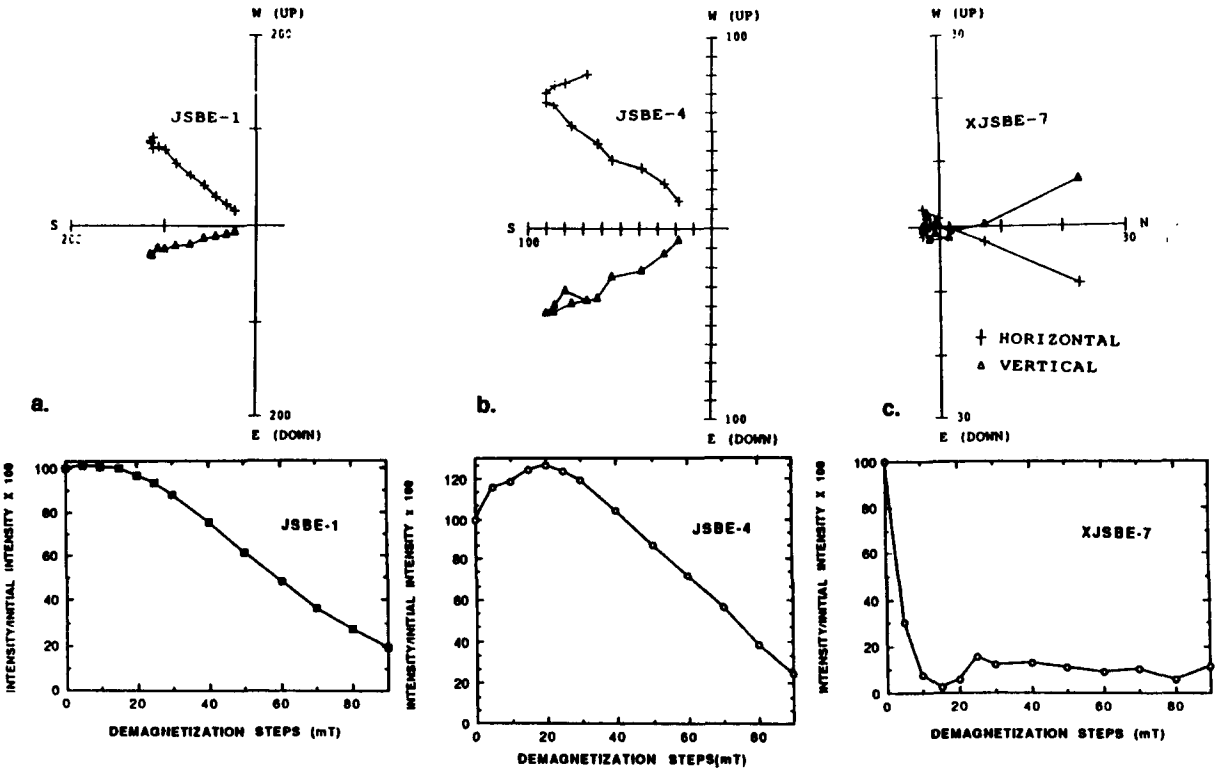


Fig. 4. Typical Zijderveld and J/Jo plots of samples from site JSBE; a.) the reversed component; b.) both the normal and the reversed components; c.) the normal component.

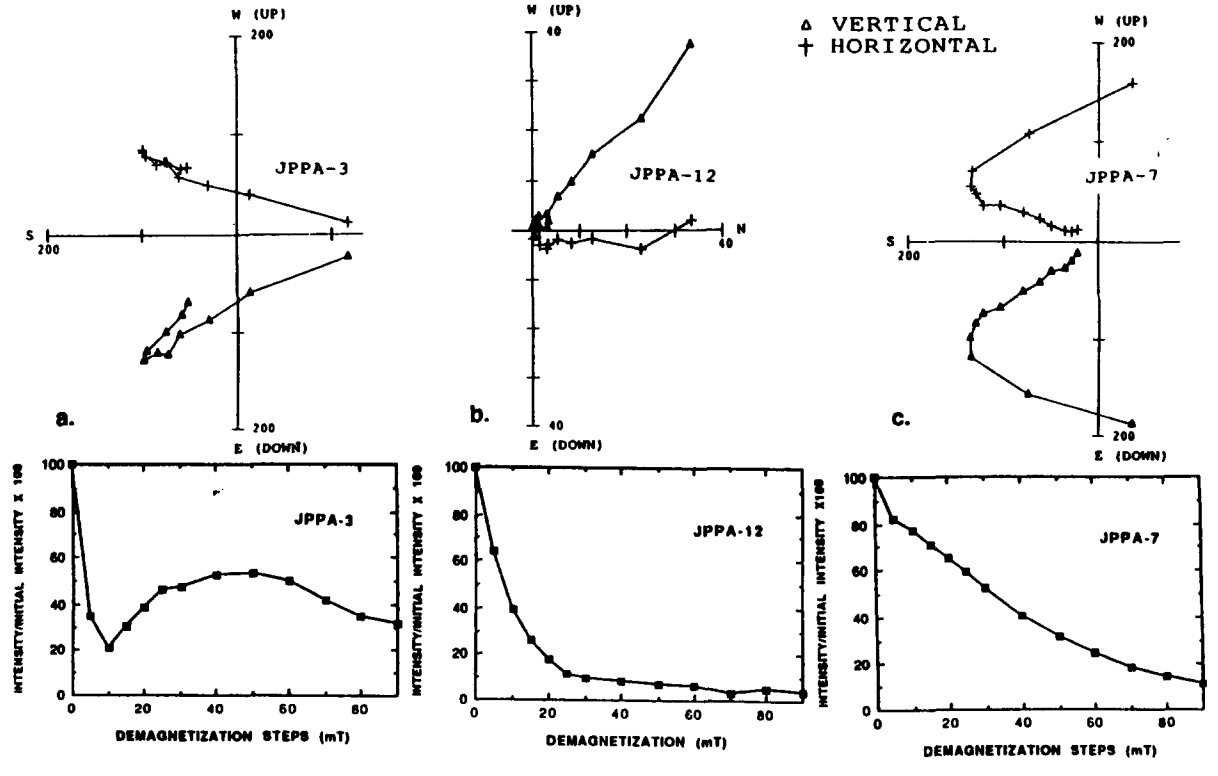


Fig. 5. Typical Zijderveld and J/Jo plots of site JPPA. a.) both normal and reversed components b.) the normal component; c.) the reversed component.

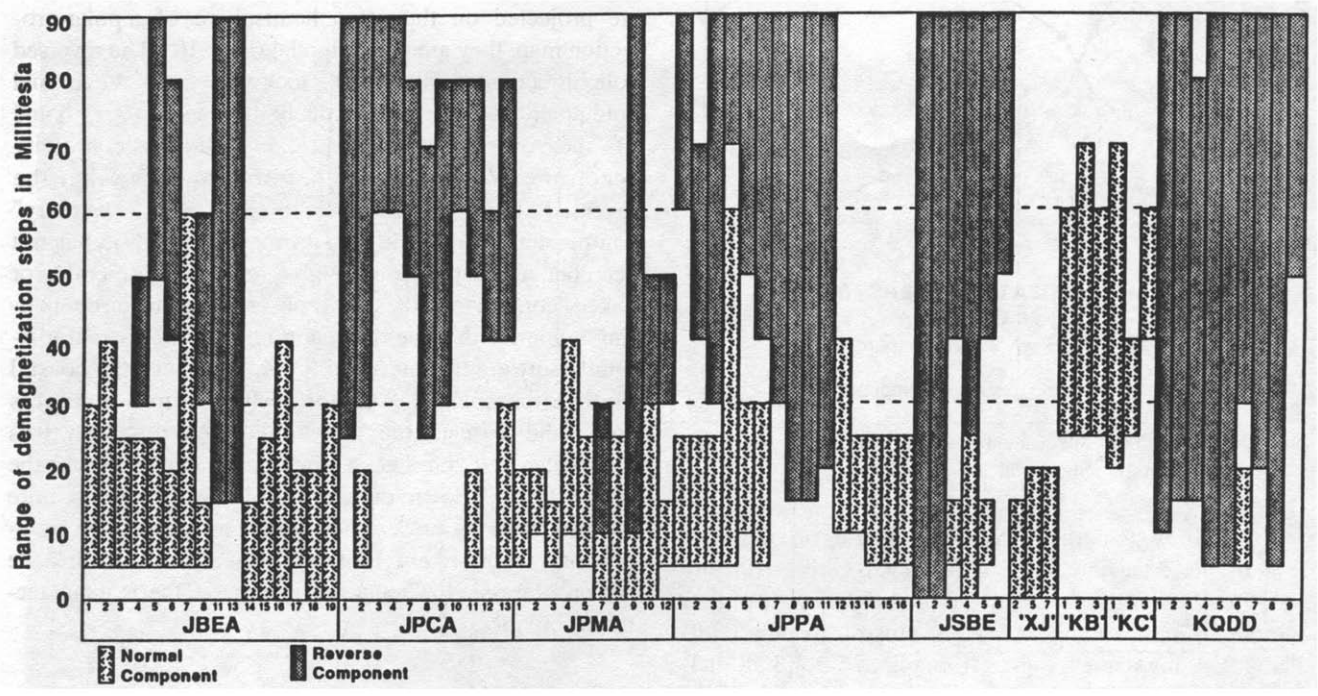


Fig. 6. Bar graph demonstrates the coercivity range of the magnetic components isolated in each of the samples that were used to calculate the site mean directions in this study.

Table 1. Paleomagnetic Results from the Coastal Plutonic Complex

ROCK UNIT	SITE	AGE	R/N*	NO.	DEC.	INC.	k	a^{95}
Granodiorite with aplite and zenoliths	JBEA	57 ±4 Rb/Sr	N	15	358.3	-49.3	35.0	6.1
			R	6	202.4	+37.8	23.5	11.8
Granite	JSBE	undated	N	6	4.8	-20.2	15.6	14.5
			R	6	216.0	+13.8	154.3	4.6
Gabbro with granitic aplite dike	JPPA	148 ±4 K/Ar	N	11	6.7	-33.9	24.9	8.5
			R	11	198.1	+34.7	41.7	6.5
Granite	JPMA	149 ±2 K/Ar	N	10	349.2	-33.2	17.8	10.5
			R	4	193.6	+45.0	17.7	16.6
Gabbro	JPCA	150 ±5 K/Ar	N	3	357.2	-41.1	9.8	25.8
			R	11	202.6	+36.6	21.2	9.2
Monzogranite	KQDB/C/D	128 ±3 K/Ar	N	7	28.2	-49.8	20.7	11.6
			R	9	229.9	+31.0	67.5	5.7

*R / N: reversed (R), or normal (N) polarity components

they typically yield the reversed direction component in the last four or five steps. These samples underwent additional thermal demagnetization from 100° to 625° C in nine steps. At 530° C they showed a slight increase in intensity and then there was a steady decay until between 600° and 625° C where there was a significant decrease in intensity (Fig. 8). The magnetic vector revealed from this demagnetization was in the same direction as the one determined from A.F. demagnetization, southerly and down. Thus in these samples it appears that reversed component is carried in part by hematite.

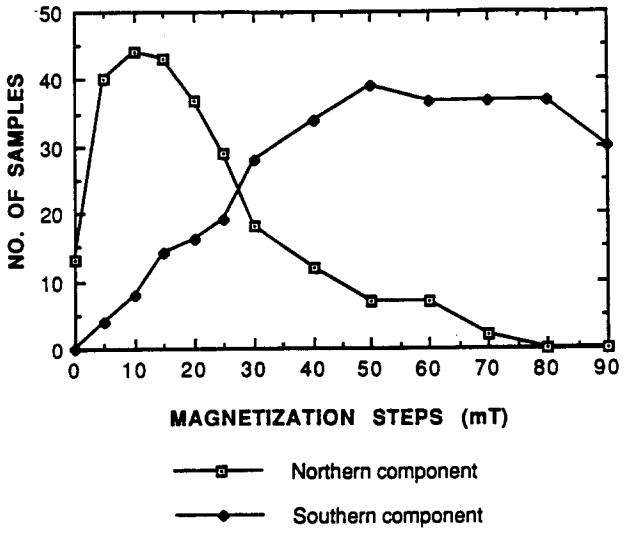


Fig. 7. The coercivity spectra showing the partial overlap of the two magnetic components isolated.

Site KQDA-D is the only sampling site on the east side of the Atacama fault that yielded cores with any statistically significant results; it is the youngest pluton sampled (Cretaceous); and it is closest to the fault. The NRM intensities range from 484.33 to 3.58 mT. Three of the eighteen samples were rejected because they had inconsistent directions. All of the rest of the samples responded well to A.F. cleaning, and all but one decayed to less than ten percent of initial intensity.

This set has both a normal and a reversed component similar to the other sites, but the coercivities of the directions overlap completely. Seven samples only have the northerly and up direction in the range of 20 to 70 mT. Nine samples have the southerly and down direction in the similar range of 5 to 90 mT. Only one of the samples has both directions, a small normal component from 5 to 20 mT. Not only is the coercivity greater for these samples' normal components, but it also is rotated clockwise from the present day magnetic field. These two factors make the normal component from this site a prime candidate for being antipodal to the reversed field instead of being a secondary present day field component (see discussion section).

In summary, six of the Jurassic to Cretaceous coastal plutons sampled for this paleomagnetic study yielded statistically significant results at the site level (see Table 1). All six sites had two characteristic directions that were isolated and defined through A.F. demagnetization, a northerly and up direction, and a southerly and down direction (Fig. 9). The normal component is very similar to the expected present day field, while the reversed component has a declination rotated significantly clockwise from the antipode of the present day field. The six normal directions were averaged to produce a regional mean with a declination of 3.4°, an inclination of -38.5°, a k value of 29.5, and an alpha 95 of 10.5°. The regional reversed mean direction had a declination of 207.8°, an inclination of 33.8°, a k value of 28.0, and an alpha 95 of 10.8°. These two mean directions were then used to calculate paleopoles. The paleopoles calculated from the normal and reversed results have the positions of latitude=85.8° N, longitude=338.2° E and latitude=63.5° S, longitude=191.9° E respectively. When the two poles are projected on the same hemisphere of a polar projection map, they are not antipodal (Fig. 10). The reversed pole direction is displaced clockwise from the normal pole position. Also, as previously discussed (Figs. 6 and 7), the reversed component has a higher coercivity spectrum. While it is not possible to argue that the reversed direction was strictly speaking "a primary" component because the actual time of aquisition has not been better constrained through a fold, conglomerate, or baked contact test, these two observations offer compelling evidence that the two components represent distinctly different times during the history of the coastal plutonic complexes. The northerly and up direction is softer and closer to the present day field direction. It is interpreted here to reflect a more recent aquisition than the southerly and down direction, which is harder and more discordant to the present day field. The latter, older component, is more likely to have been acquired during the region's Jurassic to Cretaceous magmatic, thermal and tectonic activities.

COMPARISON OF RESULTS

The results obtained in this study from the coastal plutonic complex of northern Chile are first compared to an averaged APW for South America calculated by Irving and Irving (1982). Secondly, all results obtained from

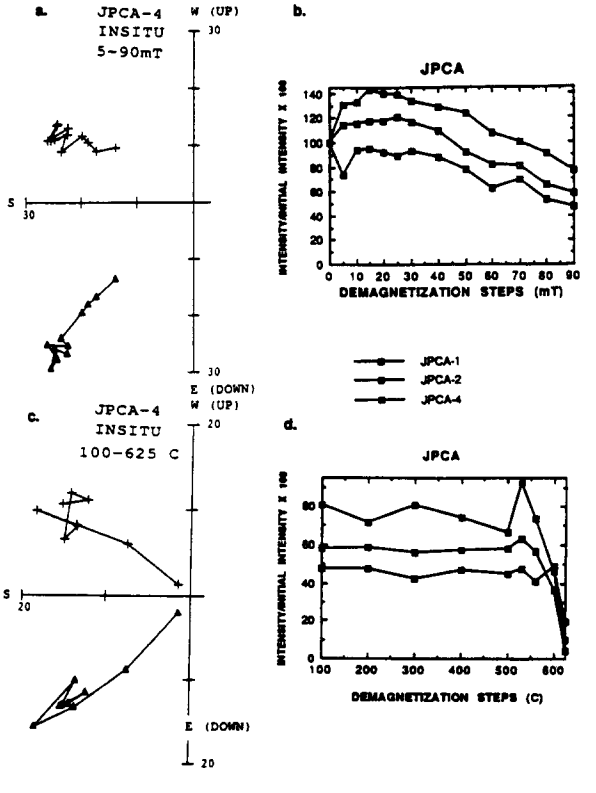


Fig. 8. Typical Zijderveld and J/Jo plots of site JPCA.

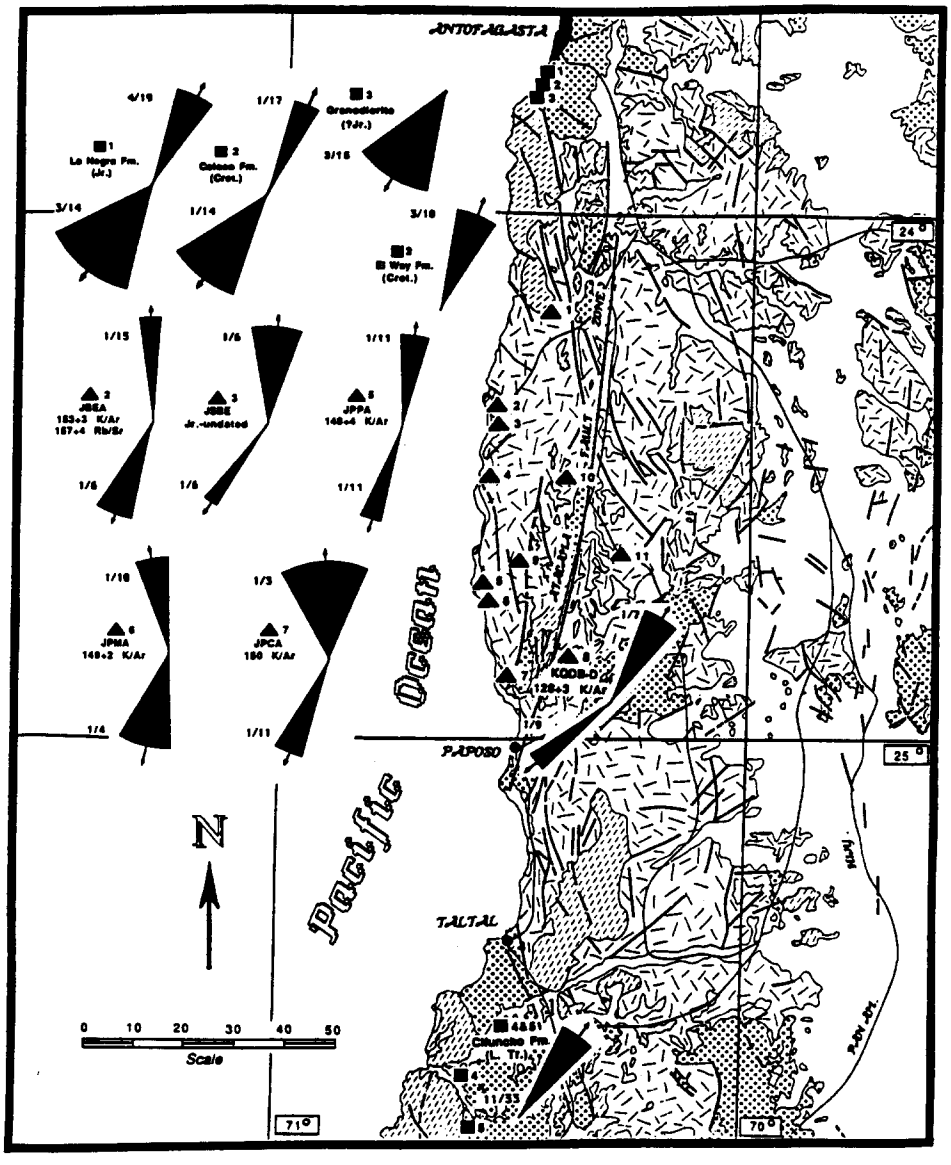


Fig. 9. Map showing the site mean declinations and alpha 95 values calculated for the plutons from this study as well those from the other published data for the region.

the sampling region are collectively compared to the same APW path. Lastly, a comparison is shown of poles obtained from coastal zones in Peru and Chile.

The Irving and Irving path was constructed by averaging individual pole determinations from 310 Ma to the present in 30 my windows to determine mean paleopoles every 10 my. From the Jurassic until the present, the calculated mean paleopoles are not dramatically different from the present dipole field. Therefore, the observed difference in declination between the normal and the reversed poles can not be easily explained by a simple variation in time of aquisition. Secular variation can also be ruled out because the six plutons were sampled over a seventy mile range, and they represent a 29 my age span.

Inclinations and declinations of the magnetic remanent magnetization to be expected without local or regional tectonism affecting the sampling sites were calculated using the running means of Irving and Irving (1982) for the Antofogasta region of Chile. For the rocks with normal polarity magnetizations acquired in the age range of 157 to 128 Ma, the declinations are expected to be near 0° (180° for reversed polarity) and inclinations between approximately 40° and 45°. Plotted along with these model directions are the observed directions for the normal and reversed site means (Fig. 11). The calculated mean normal direction of this study is more or less concordant with the expected direction within the limits of confidence, but the calculated reversed mean direction from this study has a slightly discordant inclination of -6° and a measurable discordant declination of 28°. The inclination is within the error limits, but could also be the result of an unresolved tilt correction since some gentle folding and tilting has been noted regionally (Ferraris and DiBiase, 1978). The observed discordance in declination is well beyond confidence limits and is compatible with significant clockwise rotation.

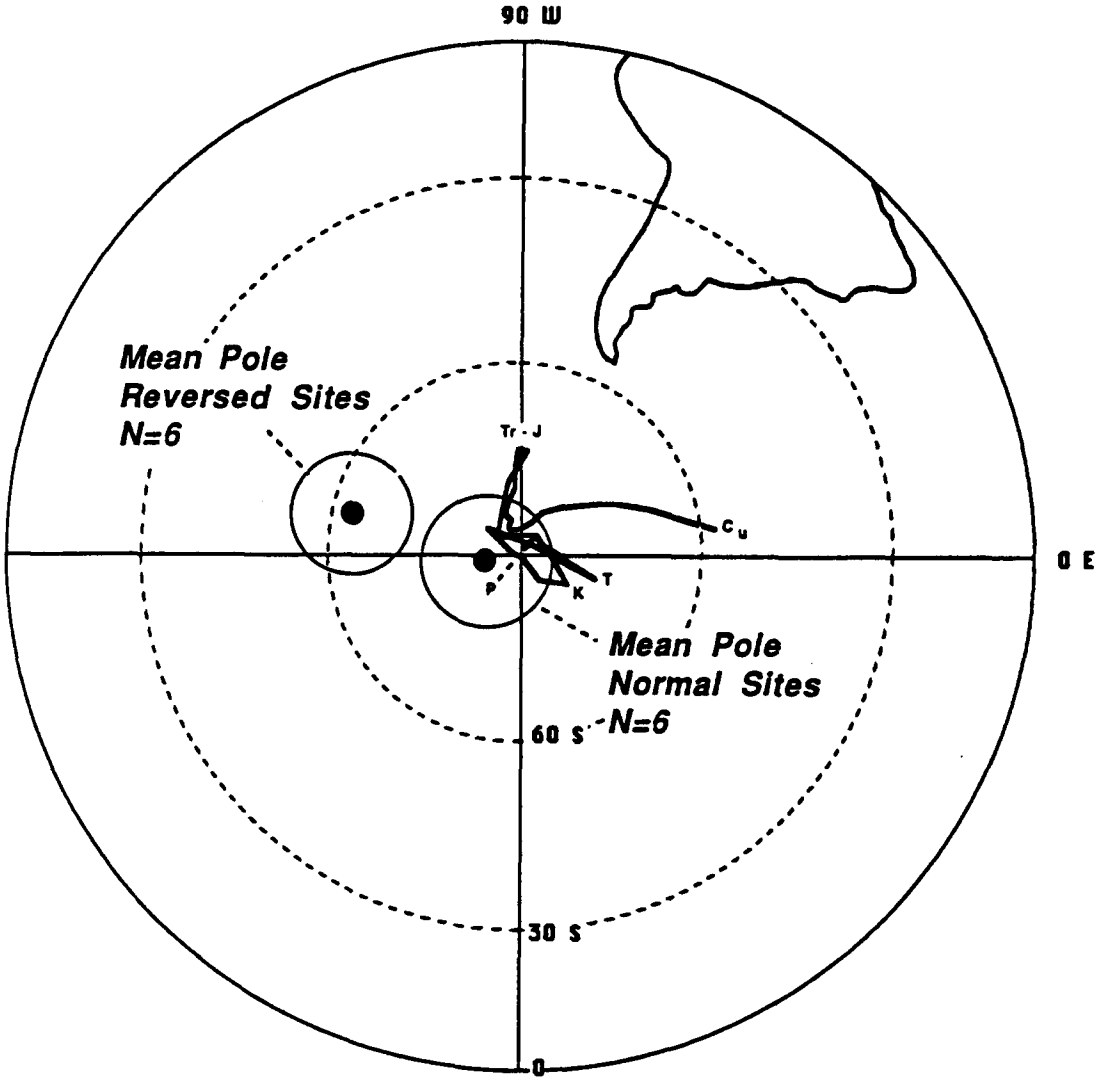


Fig. 10. Polar projection map showing an averaged APW path (Irving and Irving, 1982) together with the normal and reversed poles calculated for the plutons of this study.

There have been three other studies very near the present investigation (see Table 2) that also show evidence for clockwise rotation. The results of Turner *et al.* (1984) and Hartley *et al.* (1988) are just to the north and the results of Forsythe *et al.* (1987) are just to the south. All the regional results are compared to the model directions in Figure 12. The comparison shows there is no systematic discrepancy in mean inclination. However, the declinations show systematic discordance of 20 to 30°. The results from Hartley *et al.* (1988), for the Jurassic granodiorites and the stratified units of the La Negra Formation and El Way Formation indicate a significant clockwise rotation of -25°. The results of Forsythe *et al.* (1987) were from Triassic volcanics. Here, the normal component failed a fold test and the insitu directions suggested approximately 20°, or more, of clockwise rotation.

On a broader scale, the results from the Jurassic to Cretaceous plutons can be compared to the growing body of paleomagnetic data from the Andes of Peru to Central Chile (Table 2 and Fig. 13). The results from the region north of 19° S (Northern Chile into Peru) generally exhibit counterclockwise discordance with respect to cratonic South American APW data (Heki *et al*, 1983; May and Butler, 1985; and Beck, 1986). In the region south of 19° S, the results have the opposite sense of discordance.

While there seems to be a systematic spatial pattern of discordance, a close inspection of the "clockwise" discordant results from Chile indicate that there are major variations in the extent (0° to -36°) as well as in the likely timing of rotation (Late Jurassic to present). In the coastal regions of Chile, for example, the amount of rotation observed for Jurassic to Early Cretaceous plutonic sites varies from one coastal block to another from -0° to 30°. Results of Irwin *et al.*, 1987, from the coastal plutonic complex to the northwest of Santiago (-32° S), indicate no significant rotations had occurred in this part of the

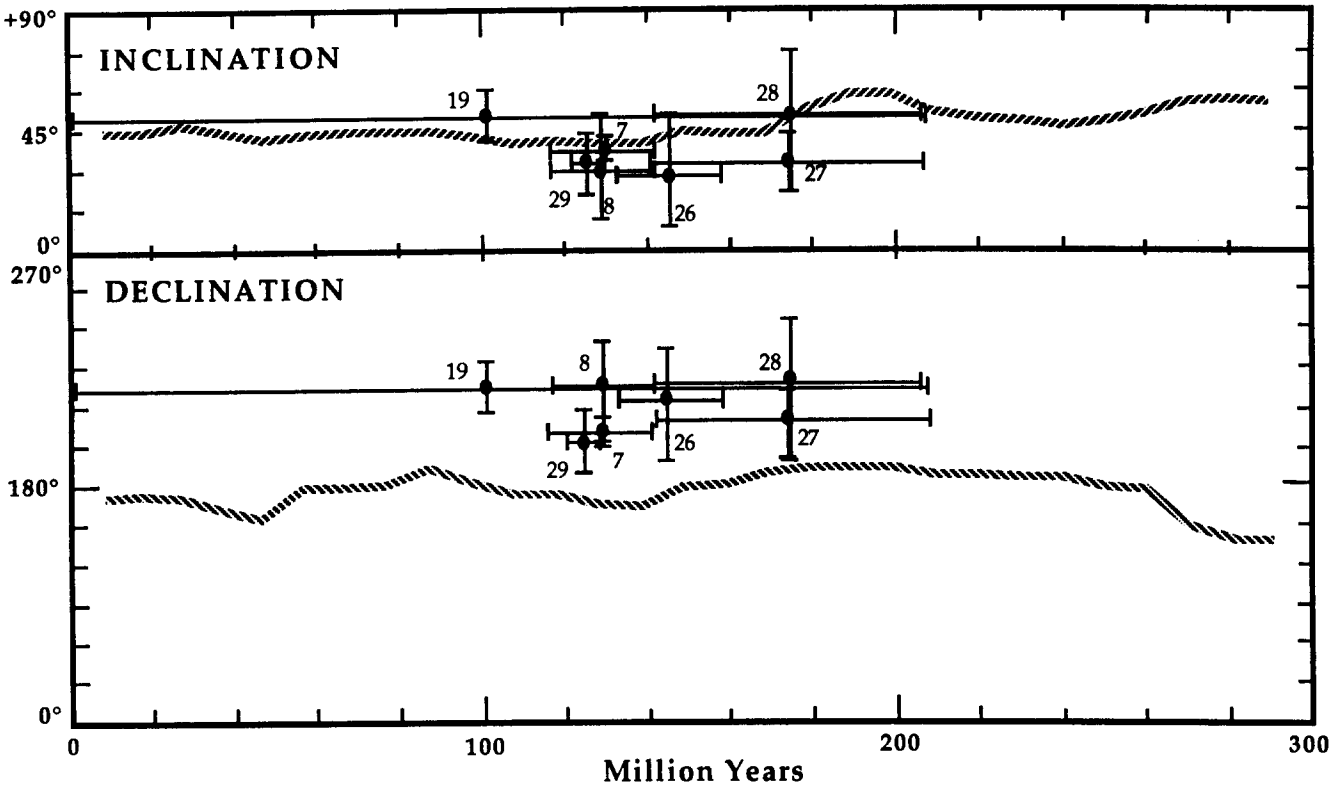


Fig. 11. The curve represents the predicted declination and inclination values calculated from the averaged APW path of Irving and Irving, 1982, for South America from 290 to 10 my B.P. The crosses represent the calculated mean declination and inclination for the present study.

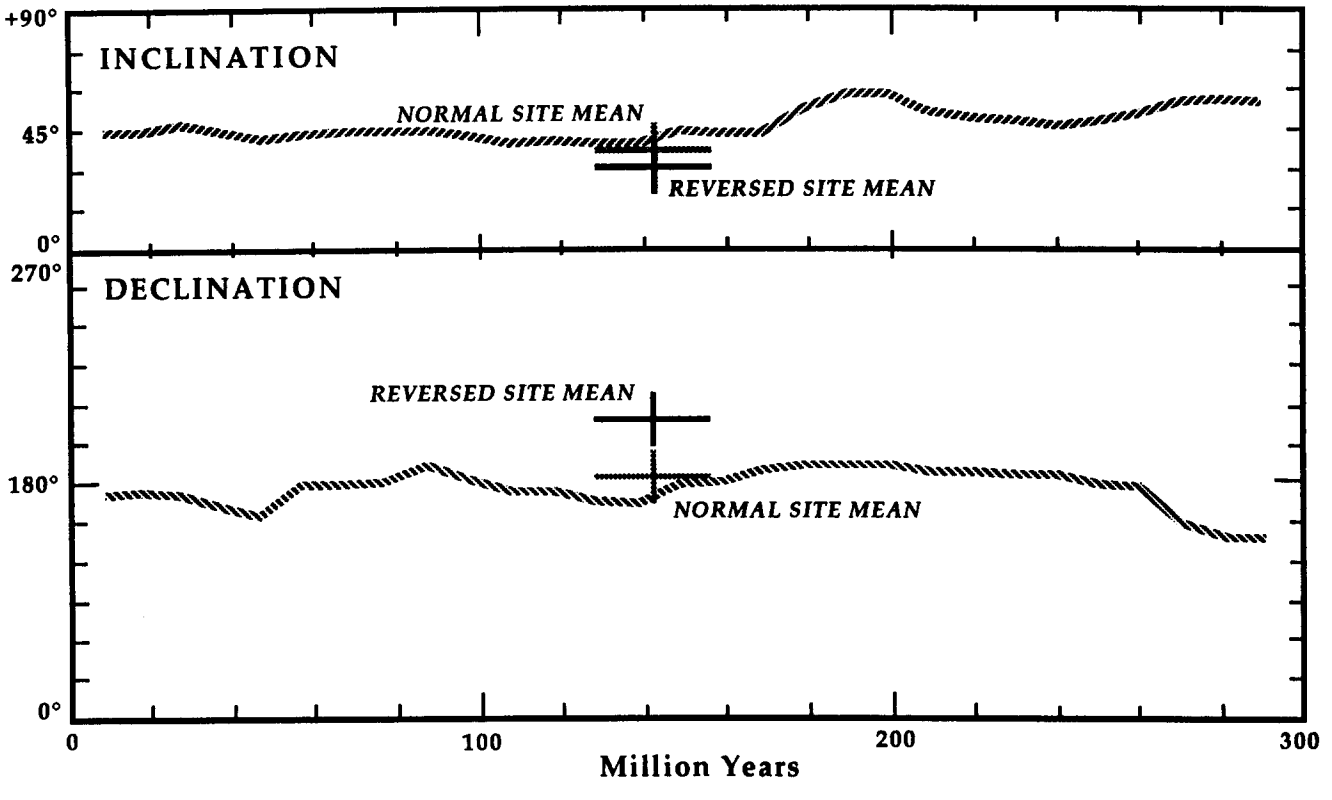


Fig. 12. Predicted versus observed declinations and inclinations for the calculated means from the regional data. Numbers refer to references listed in Table 2.

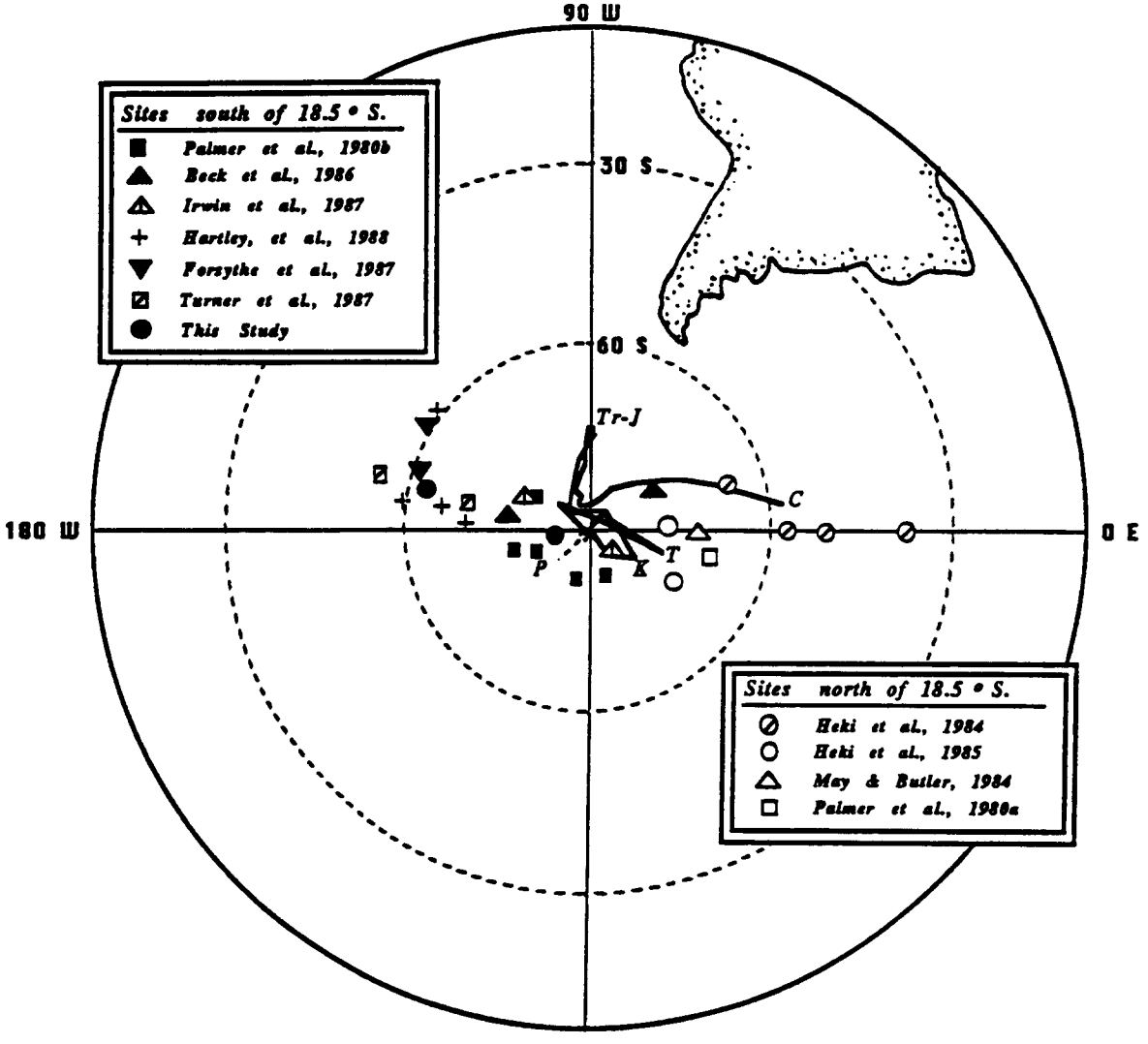


Fig. 13. Polar projection map showing the published paleomagnetic poles listed in Table 2. The solid line represents the polar wander path from 290 to 01 my B.P. as calculated by Irving and Irving, 1982.

coastal ranges. While the results reported here and previously for similar aged units to the north indicated clockwise rotations in excess of 25°, Miocene intrusions along the Liquini-Ofqui Fault Zone in southern Chile (-40°) have been found with rotations (Garcia *et al.*, 1988) comparable to that documented in this study.

The results from two paleomagnetic studies, with sampling sites in the Main Cordillera, also have yielded mixed results. Jesinkey *et al.* (1987) investigated formations that were Late Paleozoic in age from lat.=-24° S, long.=-69° W (see Table 2). The timing of the acquisition of the magnetism measured was constrained by positive conglomerate tests and by tilt (bedding) corrections. A Cretaceous section near La Serena was the subject of a study undertaken by Palmer *et al.* (1980a) (see Table 2). They found the calculated paleomagnetic poles from the section were basically concordant with the Mid- to Late Cretaceous poles derived from rock units of the stable craton of South America. The age of the magnetization component was confirmed to be pre-Tertiary in age by a positive fold test. Newer results of MacFadden and others (1990) from localities east of those sampled by Jesinkey and others (1987) in southern Bolivia suggested clockwise rotations of approximately 17.8° ± 4°. Preliminary reports by Riley and others (1990) from the central Chilean Andes also suggest clockwise rotations in the main cordillera of up to 36° (from Triassic units). Thus, at the present time, we conclude that the paleomagnetic data from the Main Cordillera in Chile and southern Bolivia indicate a complex and inconsistent pattern of rotations both north to south as well as east to west.

Table 2. Paleomagnetic Data for Andean Coastal Complexes

REFERENCE	UNIT	AGE	LONG.	LAT.	PALEOPOLE LAT./LONG.	MEAN DEC/INC	k	a^{95}	dP/dM	#SITES SAMPLES
1. Palmer *et al.*, 1980	Cameraca Fm.	157±4 Ma	70.3W	18.6S	12.9E/70.7S	339.6/-35.7	17.9	6.1	4.3/7.1	33 samples
2. Palmer *et al.*, 1980	Arqueros Fm.	Cretaceous	70.9W	29.8S	161E/80.5S	8/-41.5	40		8.4/13.2	6 sites
3. same as above	Queb. Marquesa Fm.	Cretaceous	70.9W	29.9S	168.8W/77/S	14.5/-46	63		4.9/7.5	11 sites
4. same as above	Vinita Fm.	Cretaceous	70.8W	29.8S	111W/82S	6.5/-55.5	54		6.0/8.1	13 sites
5. same as above	Elquinos Fm.	Cretaceous	70.8W	29.8S	76E/83S	356.0/-41.5	26		10.7/16.5	6 sites
6. same as above	Avg. w/o Elquinos Fm.	Cretaceous	71W	25S	209E/81S			4.5		30 sites
7. Turner *et al.*, 1984	Coloso Fm.	Early Cretaceous	70.4W	23.8S	192E/70S	21/-36	28.3	6.8		17 sites
8. same as above	Coloso Fm.	Early Cretaceous	70.4W	23.8S	195E/55S	217/+31	7.6	20.0		9 sites
9. Heki *et al.*, 1984	coastal volcanic rocks	late Middle Albian	75-80W	5-15S	0.5E/58S			5.1		12 sites
10. same as above	Chulec Fm.	Albian	78.3W	7.1S	1.6E/52.0S	321.7/-22.6	36	10.2	4.7/8.8	7 samples
11. same as above	Pariatambo Fm.	Albian	78.3W	7.1S	3.4E/38.1S	307.5/-21.1	154	5.4	3.1/5.8	6 samples
12. same as above	Yumagual Fm.	Cenomanian	78.2W	5.9S	339.7E/66.7S	339.4/-32.2	103	5.1	3.2/5.7	9 samples
13. Heki *et al.*, 1985	Atajana Fm.	Lower Cretaceous	70.3W	18.6S	37.9E/74.7S	345.1/-25.6	46	4.1	2.4/4.4	28 samples
14. same as above	Arica dike swarm	post Lower Cretaceous	70.3W	18.6S	352.4E/77.2S			3.3		19 sites
15. May & Butler, 1985	Puenta Pierda Fm.	90±5 Ma	77.1W	11.9S	359.3E/73.3S	343.2/-28.6	167.4	3.4	3.7/2.0	12 sites
16. Beck *et al.*, 1986	San Fernando - tuffs	approx. 100 Ma	71W	34S	191.6E/75.5S	16.8/-50.2	27.7	6.0		22 sites
17. same as above	Chacabuco - tuffs	approx. 65 Ma	71W	33S	330.3E/77.5S	348.9/-60.9	22.3	8.3		15 sites
18. Forsythe *et al*, 1987	Pastos Blancos Fm.	Late Carb.-E. Permian	70.1W	31.1S	200.3E/60.7S	33.1/-44.4	65	6.9	5.5/8.7	8 sites
19. same as above	Cifuncho Fm.	Late Triassic	70.6W	25.6S	217E/57.5S	36.6/-50.0	24	9.6	8.6/12.8	11 sites
20. same as above	Pichidangui Fm.	Late Triassic	70.6W	31.2S	20.7E/84.1S	353.1/-50.0	69	5.8	5.2/7.8	10 sites
21. same as above	Pichidangui Fm.	Late Triassic	71.5W	31.2S	277.5E/59.0S	191.7/+74.6	134	6.6	10.9/12	5 sites
22. Jesinkey *et al.*, 1987	La Tabla Fm.	Upper Carb. - Permian	69.3W	24.5S	347E/51S	136.9/+58.8	72.4	5.7		10 sites
23. same as above	Pular & Cas Fms.	Upper Carb. - Permian 290±7 Ma	69W	24S	350E/57S	143.8/+56.1	26.3	8.6		10 sites
24. Irvin *et al.*, 1987	dikes & plutons	approx. 170 Ma	71.7W	32-33.5S	36.7E/85.5S	2/-49.2	17.4	11.9		8 sites
25. same as above	dikes & plutons	approx. 145 Ma	71.7W	32-33.5S	207.8E/80.3S	11.4/-54.3	21.6	9.5		10 sites
26. Hartley *et al.*, 1988	Jurassic granodiorites	159 - 134 Ma	70W	24S	189E/59S	212/+28	36.0	20.8		3 sites
27. same as above	La Negra Fm.	Middle Jurassic	70W	24S	190E/66S	25/-33	70.8	11.0		4 sites
28. same as above	La Negra Fm.	Middle Jurassic	70W	24S	220E/57S	217/+50	17.2	26.4		3 sites
29. same as above	El Way Fm.	131 - 124 Ma	70W	24S	184E/70S	20/-32	117.1	11.4		3 sites

DISCUSSION OF TECTONIC SIGNIFICANCE

The combined data now make it clear that the coastal ranges of northern Chile, from approximately 26 to 23° S, have been subjected to moderate (24-32°) amounts of clockwise rotation (e.g. as shown in Fig. 12).

There are basically two sets of structural models by which to explain the observed rotations. One of these is a large regional rotation occurring in concert with oroclinal bending around the Peru to Chile bend in the Andes; along the lines initially suggested by Carey (1955), further supported by paleomagnetic studies (Heki *et al.*, 1983, 1984, 1985; MacFadden and others, 1990) and advocated by Isacks (1988) to explain late Tertiary developments of the Central Andean Plateau. The other type of model is one of localized wrench faulting and block rotation seen as occurring in association with the motions along margin-parallel fault zones such as the Atacama or Liquine-Ofqui Fault Zones. A model of this nature has been advocated by Beck (1988) and Hartley and others, (1988) in which the rotations are seen as being driven by periods of oblique subduction. An alternative driving force for the disruption of coastal blocks is the presence of a localized region of high horizontal compressive stress acting across the plate margin, such as that potentially arising from collisions of bathymetric features (Forsythe and Nelson, 1985). This type of model, essentially an "indenter" analog, has been advocated to explain late Cenozoic motions along the Liquine-Ofqui fault (Forsythe and Nelson, 1985; Nelson *et al.*, this vol.), or recognized as a potential contributing factor to localized rotations in the vicinity of the Liquine-Ofqui Fault (Garcia, 1988).

The Oroclinal Bending Hypothesis, as most recently advocated by Isacks (1988), does not in any simple fashion explain the variations in discordance that are seen east to west, and north to south in the Chilean Andes. From the coast (at latitudes of -24° S), to the high Andean Plateau and the eastern ranges in Bolivia, to the east, workers (as outlined above) have documented 30°, 0° and 17° of clockwise rotation. From north to south in the main cordillera of Chile there are similar inconsistencies (as also discussed above).

Localized block rotations. In contrast to the inconsistencies in discordance seen across, and along, the Andes at the 300 to 1000 kilometer scale, there is a high degree of consistency in the observed discordance from the northern Chilean Coast Ranges from approximately 24° to 26° S. The scale dependence of this consistency, and the proximity of the sampling localities (with discordant results) to the Atacama Fault Zone support the rotations being the product of localized block rotations that have occurred in conjunction with motion on the Atacama Fault Zone (AFZ). The main trace of the Atacama Fault Zone in the study area was shown on Figures 2 and 9. The main trace in this area has been referred to as the Paposo Fault and the Salar de Carmen Fault by Arabasz (1971) and Hervé (1987a), respectively. As discussed previously, the motion on this main fault trace has only been west side up, dip slip motion during the late Tertiary, however, regional mapping and structural analysis of mylonite zones within the Jurassic and Cretaceous plutonic units of the coastal batholith had also documented a Late Jurassic to Early Cretaceous period of sinistral movements. Recall also that an undeformed Early Cretaceous pluton cuts the mylonites and has been dated at 131 my (Hervé 1987b). Recently published results of Hervé, 1987b, indicate the existence of a synthetic (sinistral) secondary system of high angle faults that have oblique (NW-SE) trends to the main N - S trend of the Atacama Fault Zone. These secondary faults are mapped some 10 to 20 kilometers west of the main fault zone, and have 0° - 30° counter-clockwise trends to AFZ. Referred to as the Izcuna fault system, these secondary faults are seen systematically displacing the cooling margins of the Cerros Ventarros granitoid (Hervé, 1987b) which has yielded an age of 144 my.

If motion along a secondary system of faults, such as those of the Izcuna fault system, are responsible for the rotations seen in the paleomagnetic data, then one should find that restoration of displacements along these faults also restores concordance to the paleomagnetic results. This, in essence, is a structural consistency test for the block rotation model. The test is shown in Figure 14. Here it is assumed that the region to the west of the AFZ is pervasively affected by secondary faults, such as those within the Izcuna fault system, and that the average spacing, trends, and displacements seen on these faults characterize the overall secondary fault system. Also it is assumed that the secondary faults have local terminations (i.e. restricted as secondary faults to the AFZ). As such, displacements along these secondary faults results in rotations (if one keeps the main fault, AFZ, fixed in its orientation). A simple visual analog is the rotations and "fault" displacements one would observe in a row of dominos as they are allowed to fall over on a desk. The amount of rotation is strictly a function of the average fault spacing, orientation, and displacement. This relationship is visually and quantitatively shown in Figure 14. An average spacing for the Izcuna secondary faults is 1.75 km., the average fault slip is 3.43 km., and the average angle of the secondary faults to the AFZ is 20°. These values give a counterclockwise block rotation of 31°, which is consistent in sense, and amount, with the discordance seen in the paleomagnetic data.

In conclusion, the coastal plutonic complexes of Peru and Chile have been the subject of numerous paleomagnetic studies, and thus far the majority of these have revealed discordant declinations in one sense or another. Some of these results have been used to support concepts of oroclinal bending, others local block rotations. Elsewhere, rotations in coastal plutonic units have been used to support terrane or microplate motions via the operation of local Euler poles of rotation (*e.g.* Grunow *et al.*, 1991). Within the coastal plutonic complexes of northern Chile, the structural, geochronologic, and paleomagnetic data now provide for an internally consistent, and reasonably constrained, model for a syn-magmatic period of counterclockwise rotation of crustal blocks within the AFZ. Structural data supports these rotations having been produced by displacements on a systematic array of secondary syn-

thetic faults trending at low to moderate angles to the AFZ. It is perhaps important to note here that the sense of rotation of these blocks is dextral within this sinistral, secondary, synthetic fault array.

Judged by the relationships observed along the AFZ, the sense of paleomagnetic discordances observed within borderland terranes elsewhere throughout the circum-Pacific (for some examples see Beck, 1983) cannot be a reliable guide to the sense of shear or transport of a borderland terrane. It certainly would, in this case, provide an incorrect estimate of the sense of oblique subduction if one assumes that this phenomenon is at the root of the displacements and rotations observed (as suggested by papers such as those by Fitch (1972) or Jarrard (1986)).

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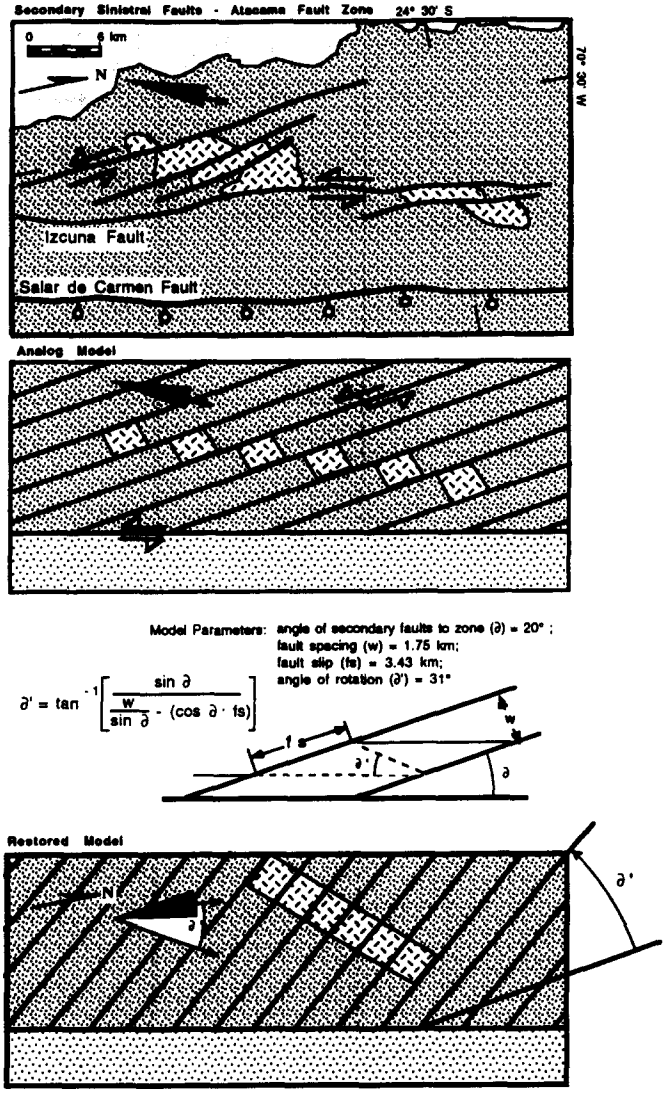


Fig. 14. Structural constraints on rotations in the Coast Ranges of northern Chile. The top panel shows the synthetic secondary fault array of the Izcuna fault system with its mapped sinistral displacements of the cooling margins of the Cerros Ventarros granitoid (144 Ma). The second panel shows the generalized version of these faults using an average spacing, orientation, and displacement derived from the upper panel. The relationships between these parameters and rotations are shown in the third panel. The bottom panel shows the restored model. On the top two the average orientation of the paleomagnetic directions are shown (in situ), and on the bottom this is restored along with the fault displacements, illustrating the consistency of these data with the structural constraints.

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